Stat 6410 Exam 1 Review

Contents

I Notation and the map of set classes **2**

II Classes of sets **2**

III Measures **4**

IV Carathéodory extension **5**

V Lebesgue–Stieltjes measures **6**

VI Completeness **8**

VII Exam question bank **9**

VII.1 The Carathéodory "steps" type question (on every past exam) 9

VII.2 Typical strategies for proofs 11

VII.3 Results proved on past exams 11

VII.4 Some potential next questions 15

VIII Examples, counterexamples and a self-check **15**

IX Appendix A tools that Chapter 1 uses **17**

I Notation and the map of set classes

$A^c = \Omega \setminus A$; $A \triangle B = (A \cap B^c) \cup (A^c \cap B)$; $\sqcup$ marks a disjoint union; $A_n \uparrow A$ means $A_n \subset A_{n+1}$ and $\bigcup_n A_n = A$ ($A_n \downarrow A$ similarly). $\sigma\langle\mathcal{A}\rangle$, $\lambda\langle\mathcal{A}\rangle$: the σ-algebra and λ-system generated by $\mathcal{A}$; $\mathcal{A}(\mathcal{C})$: the algebra generated by a semialgebra $\mathcal{C}$. μ^*: outer measure; $\mathcal{M}_{\mu^*}$: its measurable sets; m: Lebesgue measure; $\mathcal{M}_m$: Lebesgue measurable sets. m.c.f.b./m.c.f.a.: monotone continuity from below/above. In $[0,\infty]$: $x + \infty = \infty$, $0 \cdot \infty = 0$, and $\infty - \infty$ is undefined.

Class	Defining axioms	Consequences and cautions
π-system (Def. 1.1.5)	$A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$	closed under finite $\cap$
semialgebra (Def. 1.3.1)	π-system; $A \in \mathcal{C} \Rightarrow A^c = \bigsqcup_{i=1}^k B_i$ with $B_i \in \mathcal{C}$	$\emptyset \in \mathcal{C}$ if $\mathcal{C} \neq \emptyset$; $\Omega \in \mathcal{C}$ is *not* required
algebra (Def. 1.1.1)	Ω; A^c; $A \cup B$	finite $\cup$, $\cap$, $\setminus$, $\triangle$
λ-system (Def. 1.1.6)	Ω; $B \setminus A$ for $A \subset B$; increasing $\cup$	complements; countable *disjoint* $\cup$; need not be a π-system
σ-algebra (Def. 1.1.2)	Ω; A^c; countable $\cup$	countable $\cap$; is every class above

$$\sigma\text{-algebra} \Rightarrow \text{algebra} \Rightarrow \text{semialgebra} \Rightarrow \pi\text{-system}, \qquad \sigma\text{-algebra} \Rightarrow \lambda\text{-system}.$$

- σ-algebra $\iff$ π-system and λ-system (E2) $\iff$ algebra closed under increasing unions (Prop. 1.1.1).
- Semialgebra closed under finite disjoint unions $\Rightarrow$ algebra (E3).
- A union of two algebras need not be an algebra (E13). An increasing union of algebras is an algebra, but an increasing union of σ-algebras need not be a σ-algebra (E14).
- None of the forward arrows reverses; Section VIII has a counterexample for each.

II Classes of sets

Let $\Omega \neq \emptyset$ and $\mathcal{C} \subset \mathcal{P}(\Omega)$.

- **Semialgebra** (Def. 1.3.1): (i) $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$; (ii) for each $A \in \mathcal{C}$ there are disjoint $B_1, \ldots, B_k \in \mathcal{C}$, $1 \leq k < \infty$, with $A^c = \bigsqcup_{i=1}^k B_i$.
- **Algebra** (Def. 1.1.1): (a) $\Omega \in \mathcal{C}$; (b) $A \in \mathcal{C} \Rightarrow A^c \in \mathcal{C}$; (c) $A, B \in \mathcal{C} \Rightarrow A \cup B \in \mathcal{C}$. Given (a)–(b), (c) is equivalent to (c′) $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$.
- **σ-algebra** (Def. 1.1.2): an algebra with (d) $A_1, A_2, \ldots \in \mathcal{C} \Rightarrow \bigcup_{n \geq 1} A_n \in \mathcal{C}$. Then $(\Omega, \mathcal{C})$ is a *measurable space*.

Proposition 1.1.1. *$\mathcal{F}$ is a σ-algebra iff $\mathcal{F}$ is an algebra and $A_n \in \mathcal{F}$, $A_n \subset A_{n+1}$ for all n $\Rightarrow \bigcup_n A_n \in \mathcal{F}$.*

Proof. Only "if" needs work. For any $B_n \in \mathcal{F}$, the sets $A_n = \bigcup_{j \leq n} B_j$ lie in $\mathcal{F}$ (algebra), $A_n \uparrow$, and $\bigcup_n B_n = \bigcup_n A_n \in \mathcal{F}$. //

Apply. to prove a class is a σ-algebra, check "algebra + increasing unions."

Examples (Exs. 1.1.1–1.1.7, Problems 1.2–1.7)**.**

- $\Omega = \{a, b, c, d\}$: $\{\emptyset, \Omega, \{a\}\}$ is not an algebra, while $\{\emptyset, \Omega, \{a\}, \{b, c, d\}\}$ is a σ-algebra (the one generated by $\{a\}$).
- $\mathcal{P}(\Omega)$ and $\{\emptyset, \Omega\}$ are σ-algebras. On a finite Ω, every algebra is a σ-algebra.
- $\mathcal{F}_5 = \{A : A \text{ or } A^c \text{ is finite}\}$ is an algebra, not a σ-algebra when $|\Omega| = \infty$; $\mathcal{F}_6 = \{A : A \text{ or } A^c \text{ is countable}\}$ is a σ-algebra, and $\sigma\langle\mathcal{F}_5\rangle = \mathcal{F}_6$ (E17).
- Intersections of σ-algebras are σ-algebras; unions need not be (E13).
- Finite generating class $\{B_1, \ldots, B_k\}$: the generated σ-algebra consists of all unions of the atoms $\bigcap_{i=1}^k B_i^{\pm}$ (each $B_i^{\pm}$ is B_i or B_i^c), so it has at most 2^{2^k} sets (Problem 1.7).

Definition 1.1.3 (generated σ-algebra)**.** $\sigma\langle\mathcal{A}\rangle$ is the intersection of all σ-algebras on Ω that contain $\mathcal{A}$ (a nonempty family, since $\mathcal{P}(\Omega)$ is one).

Fact (rules for $\sigma\langle\cdot\rangle$)**.** (i) $\sigma\langle\mathcal{A}\rangle$ is the *smallest* σ-algebra containing $\mathcal{A}$: if $\mathcal{A} \subset \mathcal{G}$ and $\mathcal{G}$ is a σ-algebra, then $\sigma\langle\mathcal{A}\rangle \subset \mathcal{G}$. (ii) $\mathcal{A}_1 \subset \sigma\langle\mathcal{A}_2\rangle \Rightarrow \sigma\langle\mathcal{A}_1\rangle \subset \sigma\langle\mathcal{A}_2\rangle$. (iii) $\mathcal{A}$ is a σ-algebra iff $\sigma\langle\mathcal{A}\rangle = \mathcal{A}$.

Apply. to show $\sigma\langle\mathcal{A}_1\rangle = \sigma\langle\mathcal{A}_2\rangle$, prove $\mathcal{A}_1 \subset \sigma\langle\mathcal{A}_2\rangle$ *and* $\mathcal{A}_2 \subset \sigma\langle\mathcal{A}_1\rangle$ (E9; the Lemma in Section V).

Definition 1.1.4 (Borel σ-algebra)**.** $\mathcal{B}(S) = \sigma\langle\text{open subsets of } S\rangle$ for a topological (e.g. metric or Euclidean) space S.

Fact (generators of $\mathcal{B}(\mathbb{R})$; Ex. 1.1.6, Problems 1.9–1.12)**.** $\mathcal{B}(\mathbb{R})$ is also generated by: the open intervals (rational endpoints suffice); the half-lines $(-\infty, x)$ ($x \in \mathbb{Q}$ suffices); the intervals $(a, b]$; the intervals $[a, b]$; the closed sets (E9); the semialgebra $\mathcal{C}$ of (3.7). Boxes do the same in $\mathbb{R}^k$. Key step: every open set in $\mathbb{R}$ is a countable union of disjoint open intervals. Singletons do *not* generate $\mathcal{B}(\mathbb{R})$; they generate only $\mathcal{F}_6$.

Technique (good sets)**.** To show every set in $\sigma\langle\mathcal{A}\rangle$ has a property P: show that $\mathcal{G} = \{A : A \text{ has } P\}$ contains $\mathcal{A}$ and is a σ-algebra (or only a λ-system, if $\mathcal{A}$ is a π-system; Cor. 1.1.3). Then $\sigma\langle\mathcal{A}\rangle \subset \mathcal{G}$.

Definition 1.1.5, 1.1.6 (π-system; λ-system)**.** $\mathcal{C}$ is a *π-system* if $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$ (e.g. the intervals of $\mathbb{R}$). $\mathcal{L}$ is a *λ-system* if (i) $\Omega \in \mathcal{L}$; (ii) $A, B \in \mathcal{L}$, $A \subset B \Rightarrow B \setminus A \in \mathcal{L}$; (iii) $A_n \in \mathcal{L}$, $A_n \uparrow \Rightarrow \bigcup_n A_n \in \mathcal{L}$. A λ-system is closed under complements and countable disjoint unions; every σ-algebra is a λ-system.

Theorem 1.1.2 (π–λ theorem)**.** *If $\mathcal{C}$ is a π-system, then $\lambda\langle\mathcal{C}\rangle = \sigma\langle\mathcal{C}\rangle$.*

Corollary 1.1.3. *If $\mathcal{C}$ is a π-system and $\mathcal{L}$ is a λ-system with $\mathcal{C} \subset \mathcal{L}$, then $\sigma\langle\mathcal{C}\rangle \subset \mathcal{L}$.*

Idea. $\lambda\langle\mathcal{C}\rangle \subset \sigma\langle\mathcal{C}\rangle$ always. For the reverse it suffices that $\lambda\langle\mathcal{C}\rangle$ is closed under $\cap$, since a π-system that is a λ-system is a σ-algebra (E2). Show this in two "good sets" steps: first intersections with sets of $\mathcal{C}$, then with sets of $\lambda\langle\mathcal{C}\rangle$.

Apply. to extend an equality from a π-system to the whole σ-algebra (Thm. 1.2.4).

III Measures

Definition 1.2.1 (measure on an algebra)**.** Let $\mathcal{F}$ be an algebra. A set function μ on $\mathcal{F}$ is a *measure* if (a) $\mu(A) \in [0,\infty]$ for all $A \in \mathcal{F}$; (b) $\mu(\emptyset) = 0$; (c) for disjoint $A_1, A_2, \ldots \in \mathcal{F}$ with $\bigcup_n A_n \in \mathcal{F}$, $\mu(\bigcup_n A_n) = \sum_{n=1}^{\infty} \mu(A_n)$. If $\mathcal{F}$ is a σ-algebra, $(\Omega, \mathcal{F}, \mu)$ is a *measure space*.

Proposition 1.2.1 (additivity plus continuity)**.** *Let $\mu : \mathcal{F} \to [0,\infty]$ with $\mu(\emptyset) = 0$ on an algebra $\mathcal{F}$. Then μ is a measure iff it is* finitely additive *($A_1 \cap A_2 = \emptyset \Rightarrow \mu(A_1 \cup A_2) = \mu(A_1) + \mu(A_2)$) and satisfies* m.c.f.b. *($A_n \uparrow$, $\bigcup_n A_n \in \mathcal{F} \Rightarrow \mu(\bigcup_n A_n) = \lim_n \mu(A_n)$).*

Idea. ($\Rightarrow$) Disjointify, $B_n = A_n \setminus A_{n-1}$, and telescope. ($\Leftarrow$) Apply m.c.f.b. to the partial unions $\bigcup_{j \le n} A_j$.

Definition 1.2.2. μ is *finite* if $\mu(\Omega) < \infty$, a *probability measure* if $\mu(\Omega) = 1$, and *σ-finite* if $\Omega = \bigcup_{n \ge 1} A_n$ with $A_n \in \mathcal{F}$ and $\mu(A_n) < \infty$ (the A_n need not be disjoint).

Examples (Exs. 1.2.1–1.2.3)**.**

- Counting measure $\mu(A) = |A|$ on $\mathcal{P}(\Omega)$: finite iff Ω is finite; σ-finite iff Ω is countable.
- Discrete probability $P(A) = \sum_i p_i I_A(\omega_i)$ with $p_i \ge 0$, $\sum_i p_i = 1$ (additivity: swap nonnegative sums).
- Lebesgue–Stieltjes: $\mu_F((a,b]) = F(b) - F(a)$ for F nondecreasing and right-continuous; finite on bounded intervals (*Radon*), hence σ-finite.

Proposition 1.2.2. *Let μ be a measure on an algebra $\mathcal{F}$ and $A, B, A_1, \ldots, A_k \in \mathcal{F}$. (i)* Monotone*: $A \subset B \Rightarrow \mu(A) \le \mu(B)$. (ii)* Finitely subadditive*: $\mu(\bigcup_{i=1}^k A_i) \le \sum_{i=1}^k \mu(A_i)$. (iii)* Inclusion–exclusion*: if every $\mu(A_i) < \infty$,*

$$\mu\Big(\bigcup_{i=1}^{k} A_i\Big) = \sum_i \mu(A_i) - \sum_{1 \le i < j \le k} \mu(A_i \cap A_j) + \cdots + (-1)^{k-1} \mu\Big(\bigcap_{i=1}^{k} A_i\Big).$$

Useful identities: $\mu(B) = \mu(A) + \mu(B \setminus A)$ for $A \subset B$, so $\mu(B \setminus A) = \mu(B) - \mu(A)$ when $\mu(A) < \infty$; and $\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B)$ always.

Proposition 1.2.3. *Let μ be a measure on an algebra $\mathcal{F}$. (i)* m.c.f.a.*: if $A_n \downarrow$, $A = \bigcap_n A_n \in \mathcal{F}$, and $\mu(A_{n_0}) < \infty$ for some n_0, then $\mu(A_n) \to \mu(A)$. (ii)* Countable subadditivity*: if $\bigcup_n A_n \in \mathcal{F}$, then $\mu(\bigcup_n A_n) \le \sum_n \mu(A_n)$.*

Proof. (i) WLOG $\mu(A_1) < \infty$. Then $C_n = A_1 \setminus A_n \uparrow A_1 \setminus A$, so by m.c.f.b. and subtractivity, $\mu(A_1) - \mu(A_n) = \mu(C_n) \to \mu(A_1 \setminus A) = \mu(A_1) - \mu(A)$. (ii) $D_n = \bigcup_{i \le n} A_i \uparrow \bigcup_i A_i$, so by m.c.f.b. and finite subadditivity, $\mu(\bigcup_i A_i) = \lim_n \mu(D_n) \le \lim_n \sum_{i \le n} \mu(A_i) = \sum_i \mu(A_i)$. //

The finiteness in (i) cannot be dropped: counting measure on $\mathbb{N}$ with $A_n = \{j : j \ge n\}$ (E11).

Apply. null-set questions: sets that differ by a null set have equal measure (monotonicity and additivity), and a countable union of null sets is null (countable subadditivity).

Theorem 1.2.4 (uniqueness of measures)**.** *Let μ_1, μ_2 be* finite *measures on $(\Omega, \mathcal{F})$ and $\mathcal{C} \subset \mathcal{F}$ a π-system with $\sigma\langle\mathcal{C}\rangle = \mathcal{F}$. If $\mu_1 = \mu_2$ on $\mathcal{C}$ and $\mu_1(\Omega) = \mu_2(\Omega)$, then $\mu_1 = \mu_2$ on $\mathcal{F}$.*

Proof. Let $\mathcal{L} = \{A \in \mathcal{F} : \mu_1(A) = \mu_2(A)\}$. $\Omega \in \mathcal{L}$ by hypothesis. If $A \subset B$ are in $\mathcal{L}$, then $\mu_i(B \setminus A) = \mu_i(B) - \mu_i(A)$ (finiteness), so $B \setminus A \in \mathcal{L}$. If $A_n \in \mathcal{L}$ and $A_n \uparrow A$, then $\mu_1(A) = \lim \mu_1(A_n) = \lim \mu_2(A_n) = \mu_2(A)$ by m.c.f.b. So $\mathcal{L}$ is a λ-system containing the π-system $\mathcal{C}$, and Cor. 1.1.3 gives $\mathcal{F} = \sigma\langle\mathcal{C}\rangle \subset \mathcal{L}$. //

Note (σ-finite version). If also $C_n \in \mathcal{C}$, $C_n \uparrow \Omega$ with $\mu_1(C_n) = \mu_2(C_n) < \infty$, finiteness can be dropped: apply the theorem to the finite measures $\mu_i(\cdot \cap C_n)$ (they agree on $\mathcal{C}$ because $C \cap C_n \in \mathcal{C}$), then let $n \to \infty$.

Apply. every "why is the extension unique?" question, and any claim that two measures agree on $\mathcal{B}(\mathbb{R})$.

Fact (new measures from old)**.** On $(\Omega, \mathcal{F})$: $\sum_n a_n\mu_n$ is a measure for measures μ_n and $a_n \geq 0$; $A \mapsto \mu(A \cap B)$ is a measure for fixed $B \in \mathcal{F}$; restricting a measure to a sub-algebra gives a measure on it.

IV Carathéodory extension

The semialgebra definition (Def. 1.3.1) is in the box of Section II. If $\mathcal{C} \neq \emptyset$, then $\emptyset \in \mathcal{C}$, and $\Omega = A \sqcup B_1 \sqcup \cdots \sqcup B_k$ is a finite disjoint union of members of $\mathcal{C}$, though $\Omega \in \mathcal{C}$ is not required. Examples 1.3.1–1.3.3: $\{(a, b], (b, \infty) : -\infty \leq a, b < \infty\}$ on $\mathbb{R}$; all intervals of $\mathbb{R}$; all boxes in $\mathbb{R}^k$. Every algebra is a semialgebra.

Problem 1.19. $\mathcal{A}(\mathcal{C}) = \{\bigsqcup_{i=1}^k B_i : B_i \in \mathcal{C} \text{ disjoint}, \ k \in \mathbb{N}\}$ is the smallest algebra containing $\mathcal{C}$, and $\sigma\langle\mathcal{A}(\mathcal{C})\rangle = \sigma\langle\mathcal{C}\rangle$.

Definition 1.3.2 (measure on a semialgebra)**.** $\mu : \mathcal{C} \to [0, \infty]$ with $\mu(\emptyset) = 0$ and $\mu(\bigsqcup_n A_n) = \sum_n \mu(A_n)$ whenever the $A_n \in \mathcal{C}$ are disjoint and $\bigsqcup_n A_n \in \mathcal{C}$.

Proposition 1.3.1. *$\bar{\mu}(\bigsqcup_{i=1}^k B_i) := \sum_{i=1}^k \mu(B_i)$ is well defined and is a measure on the algebra $\mathcal{A}(\mathcal{C})$ extending μ.*

Definition 1.3.3, 1.3.5 (outer measure)**.** The *outer measure induced by* μ is, for every $A \subset \Omega$,

$$\mu^*(A) = \inf\Big\{\sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n\geq 1} \subset \mathcal{C}, \ A \subset \bigcup_{n\geq 1} A_n\Big\}. \tag{3.1}$$

In general, an *outer measure* is any $\mu^* : \mathcal{P}(\Omega) \to [0, \infty]$ with (3.4) $\mu^*(\emptyset) = 0$; (3.5) $A \subset B \Rightarrow \mu^*(A) \leq \mu^*(B)$; (3.6) $\mu^*(\bigcup_n A_n) \leq \sum_n \mu^*(A_n)$.

Definition 1.3.4 (μ^*-measurable; Carathéodory condition)**.** A is μ^*-measurable if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \qquad \text{for all } E \subset \Omega. \tag{3.3}$$

"$\leq$" in (3.3) always holds by subadditivity, so checking measurability means checking $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ (only E with $\mu^*(E) < \infty$ matter).

Theorem 1.3.2 (Carathéodory)**.** *Let μ^* be an outer measure on Ω and $\mathcal{M} = \mathcal{M}_{\mu^*}$ the μ^*-measurable sets. Then (i) $\mathcal{M}$ is a σ-algebra; (ii) μ^* restricted to $\mathcal{M}$ is a measure; (iii) $\mu^*(A) = 0 \Rightarrow \mathcal{P}(A) \subset \mathcal{M}$.*

Idea. (i)–(ii): (3.3) is symmetric in A, A^c; finite unions by splitting E into four pieces; countable unions by disjointifying and using $\mu^*(E \cap \bigsqcup_{j\leq n} B_j) = \sum_{j\leq n} \mu^*(E \cap B_j)$ for disjoint $B_j \in \mathcal{M}$.

Proof of (iii). Let $\mu^*(A) = 0$ and $B \subset A$. For any E: $\mu^*(E \cap B) \le \mu^*(A) = 0$ and $\mu^*(E \cap B^c) \le \mu^*(E)$, so $\mu^*(E \cap B) + \mu^*(E \cap B^c) \le \mu^*(E)$; "$\ge$" is subadditivity. Hence $B \in \mathcal{M}$. //

Definition 1.3.6 (complete measure space)**.** $(\Omega, \mathcal{F}, \nu)$ is *complete* if $A \in \mathcal{F}$ and $\nu(A) = 0$ imply $\mathcal{P}(A) \subset \mathcal{F}$. By Thm. 1.3.2(iii), $(\Omega, \mathcal{M}_{\mu^*}, \mu^*)$ is complete.

Theorem 1.3.3 (Carathéodory's extension theorem)**.** *Let μ be a measure on a semialgebra $\mathcal{C}$ and μ^* as in (3.1). Then (i) μ^* is an outer measure; (ii) $\mathcal{C} \subset \mathcal{M}_{\mu^*}$; (iii) $\mu^* = \mu$ on $\mathcal{C}$.*

Idea. (i) $\epsilon/2^n$ covers (Problem 1.20). (ii) For $A \in \mathcal{C}$, a $\mathcal{C}$-cover of E splits into $\mathcal{C}$-covers of $E \cap A$ and $E \cap A^c$. (iii) "$\le$": A covers itself; "$\ge$" uses countable additivity of μ on $\mathcal{C}$.

$\mathcal{C}$	$\subset$	$\mathcal{A}(\mathcal{C})$	$\subset$	$\sigma\langle\mathcal{C}\rangle$	$\subset$	$\mathcal{M}_{\mu^*}$	$\subset$	$\mathcal{P}(\Omega)$
μ		$\bar{\mu}$		$\mu^*\|_{\sigma\langle\mathcal{C}\rangle}$		$\mu^*\|_{\mathcal{M}_{\mu^*}}$		μ^*
measure		measure (1.3.1)		measure		complete measure		outer measure only

Since $\mathcal{M}_{\mu^*}$ is a σ-algebra containing $\mathcal{C}$, $\sigma\langle\mathcal{C}\rangle \subset \mathcal{M}_{\mu^*}$, so $(\Omega, \sigma\langle\mathcal{C}\rangle, \mu^*)$ is also a measure space, though possibly incomplete (Section VI).

Apply. Steps 4–7 of the Carathéodory template (Section VII.1).

Theorem 1.3.6 (uniqueness of the extension)**.** *Let μ be a σ-finite measure on a semialgebra $\mathcal{C}$ and ν a measure on $(\Omega, \sigma\langle\mathcal{C}\rangle)$ with $\nu = \mu$ on $\mathcal{C}$. Then $\nu = \mu^*$ on $\sigma\langle\mathcal{C}\rangle$.*

Idea. Apply Thm. 1.2.4 to $\nu(\cdot \cap \Omega_n)$ and $\mu^*(\cdot \cap \Omega_n)$ on disjoint pieces $\Omega_n \in \mathcal{C}$ of finite measure, then sum over n (Problem 1.26).

Apply. uniqueness when the measure is infinite but σ-finite, e.g. Lebesgue measure.

Also in §1.3.4 (statements only). **Thm. 1.3.4:** if $A \in \mathcal{M}_{\mu^*}$ and $\mu^*(A) < \infty$, then for every $\epsilon > 0$ there are disjoint $B_1, \dots, B_k \in \mathcal{C}$ with $\mu^*(A \triangle \bigsqcup_{j=1}^k B_j) < \epsilon$. **Cor. 1.3.5 (regularity):** if μ is Radon on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ and $\mu(A) < \infty$, then for every $\epsilon > 0$ there are K compact and G open with $K \subset A \subset G$ and $\mu(G \setminus K) < \epsilon$.

V Lebesgue–Stieltjes measures

Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing, with one-sided limits $F(x\pm)$ and $F(\pm\infty) = \lim_{x \to \pm\infty} F(x)$. Set

$$\mathcal{C} = \{(a, b] : -\infty \le a \le b < \infty\} \cup \{(a, \infty) : -\infty \le a < \infty\}, \tag{3.7}$$

$$\mu_F((a, b]) = F(b+) - F(a+), \qquad \mu_F((a, \infty)) = F(\infty) - F(a+). \tag{3.8}$$

For right-continuous F (e.g. a cdf): $\mu_F((a, b]) = F(b) - F(a)$ and $\mu_F((a, \infty)) = F(\infty) - F(a)$.

Lemma (the three facts every "steps" answer cites)**.** *(i) $\mathcal{C}$ is a semialgebra; (ii) $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$; (iii) μ_F is a measure on $\mathcal{C}$ (Problems 1.22–1.23).*

Sketch. (i) compute intersections and complements; (ii) two containments via interval identities; (iii) cite Heine–Borel.

Proof. (i) $(a,b]\cap(c,d]=(a\vee c,\,b\wedge d]$, $(a,b]\cap(c,\infty)=(a\vee c,\,b]$, $(a,\infty)\cap(c,\infty)=(a\vee c,\infty)$, read as $\emptyset=(x,x]\in\mathcal{C}$ when empty. Complements: $(a,b]^c=(-\infty,a]\sqcup(b,\infty)$ (just (b,∞) if $a=-\infty$), $(a,\infty)^c=(-\infty,a]$, and $\emptyset^c=(-\infty,\infty)\in\mathcal{C}$.

(ii) $(\subset)$ $(a,b]=\bigcap_n(a,b+\frac{1}{n})$ and (a,∞) is open, so $\mathcal{C}\subset\mathcal{B}(\mathbb{R})$ and $\sigma\langle\mathcal{C}\rangle\subset\mathcal{B}(\mathbb{R})$. $(\supset)$ $(a,b)=\bigcup_n(a,b-\frac{1}{n}]\in\sigma\langle\mathcal{C}\rangle$, and every open set is a countable union of open intervals, so $\mathcal{B}(\mathbb{R})\subset\sigma\langle\mathcal{C}\rangle$.

(iii) Countable additivity on $\mathcal{C}$ is the only nontrivial input: reduce to right-continuous F and use compactness of $[c,b]$ (Heine–Borel) to compare $F(b)-F(a)$ with $\sum_n(F(b_n)-F(a_n))$ when $(a,b]=\bigsqcup_n(a_n,b_n]$. //

Definition 1.3.7, 1.3.8 (Lebesgue–Stieltjes and Lebesgue measure)**.** The Carathéodory extension $(\mathbb{R},\mathcal{M}_{\mu_F^*},\mu_F^*)$ of μ_F is the *Lebesgue–Stieltjes measure space*; since $\sigma\langle\mathcal{C}\rangle=\mathcal{B}(\mathbb{R})\subset\mathcal{M}_{\mu_F^*}$, μ_F^* is a measure on $(\mathbb{R},\mathcal{B}(\mathbb{R}))$, finite on bounded intervals. For $F(x)=x$, $m=\mu_F^*$ is *Lebesgue measure* and $\mathcal{M}_m=\mathcal{M}_{\mu_F^*}$ the *Lebesgue measurable sets*.

Fact (Radon = Lebesgue–Stieltjes; cdfs)**.** A measure μ on $(\mathbb{R},\mathcal{B}(\mathbb{R}))$ that is finite on bounded intervals equals μ_F^* for $F(x)=\mu((0,x])$ $(x>0)$, $F(0)=0$, $F(x)=-\mu((x,0])$ $(x<0)$. In particular a probability measure P on $\mathcal{B}(\mathbb{R})$ is determined by its cdf $F(x)=P((-\infty,x])$. F, $F+c$ and $F(\cdot+)$ all generate the same measure.

Fact (Lebesgue measure; §1.3.2 (i)–(v))**.**

(i) $\mu_F^*(\{x\})=F(x+)-F(x-)$ (the jump of F at x), so $m(\{x\})=0$.
(ii) Countable sets are m-null (e.g. $m(\mathbb{Q})=0$).
(iii) The Cantor set C is uncountable yet $m(C)=0$: the removed middle thirds have total length $\sum_{n\geq1}2^{n-1}3^{-n}=1$.
(iv) $m(E+c)=m(E)$, $m(-E)=m(E)$, and $m(cE)=|c|\,m(E)$ for $E\in\mathcal{B}(\mathbb{R})$, $c\in\mathbb{R}$.
(v) Non-Lebesgue-measurable sets exist (Vitali's construction, using the axiom of choice).

Fact (μ_F^* of intervals, F nondecreasing, $a<b$)**.** These follow from (3.8), $\{x\}=\bigcap_n(x-\frac{1}{n},x]$ and m.c.f.a.

Set	μ_F^*	Set	μ_F^*
$\{a\}$	$F(a+)-F(a-)$	(a,∞)	$F(\infty)-F(a+)$
$(a,b]$	$F(b+)-F(a+)$	$[a,\infty)$	$F(\infty)-F(a-)$
$[a,b]$	$F(b+)-F(a-)$	$(-\infty,b]$	$F(b+)-F(-\infty)$
(a,b)	$F(b-)-F(a+)$	$(-\infty,b)$	$F(b-)-F(-\infty)$
$[a,b)$	$F(b-)-F(a-)$	$\mathbb{R}$	$F(\infty)-F(-\infty)$

If F is continuous (e.g. the Exponential and Normal cdfs), open versus closed endpoints do not matter.

On $\mathbb{R}^2$ (§1.3.3): F must satisfy the rectangle inequality $F(a_2,b_2)-F(a_2,b_1)-F(a_1,b_2)+F(a_1,b_1)\geq0$ and $F(a_2,b_2)-F(a_1,b_1)\geq0$ for $a_1\leq a_2$, $b_1\leq b_2$; then $\mu_F((a_1,a_2]\times(b_1,b_2])=F(a_2+,b_2+)-F(a_2+,b_1+)-F(a_1+,b_2+)+F(a_1+,b_1+)$ (3.12), and Carathéodory applies as on $\mathbb{R}$. $F(a,b)=ab$ gives Lebesgue measure (area) on $\mathbb{R}^2$.

VI Completeness

Recall (Def. 1.3.6): $(\Omega, \mathcal{F}, \mu)$ is complete if every subset of a μ-null set in $\mathcal{F}$ is in $\mathcal{F}$.

- $(\mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$ is complete (Thm. 1.3.2(iii)); $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F^*)$ need not be.
- $(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$ is not complete: every subset of the Cantor set is Lebesgue measurable, and there are more such subsets ($2^{\mathfrak{c}}$) than Borel sets ($\mathfrak{c}$).
- Discrete F (e.g. a point mass): $\mathcal{M}_{\mu_F^*} = \mathcal{P}(\mathbb{R})$ (Problem 1.28), yet $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F^*)$ is not complete.

Theorem 1.4.1 (completion)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $\tilde{\mathcal{F}} = \{A : B_1 \subset A \subset B_2$ for some $B_1, B_2 \in \mathcal{F}$ with $\mu(B_2 \setminus B_1) = 0\}$, and $\tilde{\mu}(A) = \mu(B_1) = \mu(B_2)$ for any such pair. Then (i) $\tilde{\mathcal{F}}$ is a σ-algebra containing $\mathcal{F}$; (ii) $\tilde{\mu}$ is well defined; (iii) $(\Omega, \tilde{\mathcal{F}}, \tilde{\mu})$ is complete and $\tilde{\mu} = \mu$ on $\mathcal{F}$.*

Idea. Complements flip the sandwich ($B_2^c \subset A^c \subset B_1^c$); countable unions sandwich between $\bigcup_n B_{1n}$ and $\bigcup_n B_{2n}$, whose difference is inside the null set $\bigcup_n (B_{2n} \setminus B_{1n})$.

Equivalently, $\tilde{\mathcal{F}} = \{B \cup N : B \in \mathcal{F},\ N \subset N_0$ for some $N_0 \in \mathcal{F}$ with $\mu(N_0) = 0\}$ and $\tilde{\mu}(B \cup N) = \mu(B)$. If μ is σ-finite on a semialgebra $\mathcal{C}$, the completion of $(\Omega, \sigma\langle\mathcal{C}\rangle, \mu^*)$ is $(\Omega, \mathcal{M}_{\mu^*}, \mu^*)$ (Problem 1.31); so $\mathcal{M}_m$ is the completion of $\mathcal{B}(\mathbb{R})$ under m.

VII Exam question bank

Scope. Chapter 1 only (§§1.1–1.4).

Question type
Define semialgebra, algebra, σ-algebra
Closure arguments for classes of sets
Trace σ-algebra
Verify that a set function is a measure
Null and full-measure sets
Carathéodory steps and uniqueness (Section VII.1)
Compute μ_F^* of a set (Section VII.1)

VII.1 The Carathéodory "steps" type question (on every past exam)

Ghosh's instruction each time: *"Write down the steps of the theorems carefully and state the results used in each step"*, so cite a result at every step.

Model answer: extend μ_F to the L–S measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

Let F be a cdf (nondecreasing, right-continuous, $F(-\infty) = 0$, $F(\infty) = 1$).

Step 1. *Semialgebra.* The class

$$\mathcal{C} = \{(a, b] : -\infty \le a \le b < \infty\} \cup \{(a, \infty) : -\infty \le a < \infty\}$$

is a semialgebra with $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$ (Def. 1.3.1; Lemma in Section V).

Step 2. *Measure on $\mathcal{C}$.* $\mu_F((a, b]) = F(b) - F(a)$ and $\mu_F((a, \infty)) = 1 - F(a)$. Then $\mu_F(\emptyset) = 0$ and μ_F is countably additive on $\mathcal{C}$ (Heine–Borel; Problems 1.22–1.23), so it is a measure on $\mathcal{C}$ (Def. 1.3.2).

Step 3. *(Optional) algebra.* $\bar{\mu}_F(\bigsqcup_i B_i) = \sum_i \mu_F(B_i)$ is a measure on $\mathcal{A}(\mathcal{C})$ (Prop. 1.3.1).

Step 4. *Outer measure.* $\mu_F^*(A) = \inf\{\sum_n \mu_F(A_n) : A \subset \bigcup_n A_n,\ A_n \in \mathcal{C}\}$ for $A \subset \mathbb{R}$ is an outer measure (Thm. 1.3.3(i)).

Step 5. *Measurable sets.* $\mathcal{M}_{\mu_F^*} = \{A \subset \mathbb{R} : \mu_F^*(E) = \mu_F^*(E \cap A) + \mu_F^*(E \cap A^c)\ \forall E \subset \mathbb{R}\}$ is a σ-algebra, and μ_F^* restricted to $\mathcal{M}_{\mu_F^*}$ is a complete measure (Thm. 1.3.2).

Step 6. *Extension.* $\mathcal{C} \subset \mathcal{M}_{\mu_F^*}$ and $\mu_F^* = \mu_F$ on $\mathcal{C}$ (Thm. 1.3.3(ii)–(iii)).

Step 7. *Restrict to Borel sets.* $\mathcal{B}(\mathbb{R}) = \sigma\langle\mathcal{C}\rangle \subset \mathcal{M}_{\mu_F^*}$ by minimality of $\sigma\langle\cdot\rangle$, so $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F^*)$ is a measure space and μ_F^* extends μ_F.

Step 8. *Uniqueness.* $\mathcal{C}$ is a π-system with $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$ and $\mathbb{R} \in \mathcal{C}$, and $\mu_F^*(\mathbb{R}) = F(\infty) - F(-\infty) = 1 < \infty$. If ν is any measure on $\mathcal{B}(\mathbb{R})$ with $\nu = \mu_F$ on $\mathcal{C}$, then $\nu(\mathbb{R}) = 1$ and $\nu = \mu_F^*$ on $\mathcal{B}(\mathbb{R})$ by Thm. 1.2.4. (For an infinite but σ-finite μ_F, such as Lebesgue measure, cite Thm. 1.3.6.)

Exponential(1) (2016 #4, 2020 #3).
$F(x) = (1 - e^{-x})I_{(0,\infty)}(x)$; the 2016 version, $(1 - e^{-x})I_{\{x \geq 0\}}$, is the same function. F is continuous, and with $x^+ = \max(x, 0)$,

$$\mu_F((a, b]) = e^{-a^+} - e^{-b^+}, \qquad \mu_F((a, \infty)) = e^{-a^+}.$$

This is the Exponential(1) probability measure, unique by Thm. 1.2.4 since $\mu_F^*(\mathbb{R}) = 1$.
For 2020 #3(c), $\mu_F^*((0, 1)) = F(1-) - F(0+) = 1 - e^{-1} \approx 0.6321$.

Standard normal (2024 #2 aka Practice Exam 1 #2).
$F = \Phi$ is continuous with $\Phi(-x) = 1 - \Phi(x)$; the measure is the standard Gaussian, $\mathrm{N}(0, 1)$, probability measure. For #2(c),

$$\begin{aligned}\mu_\Phi^*(\{x : |x| \geq 0.5\}) = \mu_\Phi^*((-\infty, -0.5]) + \mu_\Phi^*([0.5, \infty)) &= \Phi(-0.5) + 1 - \Phi(0.5) \\ &= 2\big(1 - \Phi(0.5)\big) = 2(1 - 0.6915) = 0.6170,\end{aligned}$$

reading $\Phi(0.5) = .6915$ from row 0.5, column .00 of the z-table.
Likewise $\mu_\Phi^*([-1.07, 1.07]) = 2\Phi(1.07) - 1 = 2(.8577) - 1 = 0.7154$.

Cubic cdf (2021 #3 = Practice Exam 2 #3).
$F(x) = x^3 I_{(0,1)}(x) + I_{[1,\infty)}(x)$ is 0 on $(-\infty, 0]$, x^3 on $(0, 1)$ and 1 on $[1, \infty)$. It is nondecreasing and continuous ($F(0+) = 0 = F(0)$, $F(1-) = 1 = F(1)$), with $F(-\infty) = 0$ and $F(\infty) = 1$, so it is a cdf. Writing $F(x) = (x^+ \wedge 1)^3$,

$$\mu_F((a, b]) = (b^+ \wedge 1)^3 - (a^+ \wedge 1)^3, \qquad \mu_F((a, \infty)) = 1 - (a^+ \wedge 1)^3.$$

Uniqueness is again Thm. 1.2.4, since $\mu_F^*(\mathbb{R}) = 1$. For #3(c), $\mu_F^*((0.5, 2)) = F(2-) - F(0.5+) = 1 - 0.5^3 = 0.875$.

Lebesgue ($F(x) = x$). $\mu_F((a, b]) = b - a$ and $\mu_F((a, \infty)) = \infty$. Here m is not finite but is σ-finite on $\mathcal{C}$ via $(n, n + 1]$, $n \in \mathbb{Z}$, so uniqueness comes from Thm. 1.3.6.

Note (slips to avoid). Use half-open $(a, b]$, not $[a, b]$ or (a, b): complements must be finite disjoint unions from $\mathcal{C}$. Keep $\emptyset$ and $\mathbb{R}$ in $\mathcal{C}$ by allowing $a \leq b$ and $a = -\infty$; the 2021 key's $a < b$ drops $\emptyset = (0, 1] \cap (2, 3]$, so its class is not closed under $\cap$. For uniqueness, name both the π-system and the finiteness (or σ-finiteness) hypothesis.

VII.2 Typical strategies for proofs

T1. Two containments. $\sigma\langle\mathcal{A}_1\rangle = \sigma\langle\mathcal{A}_2\rangle$ via $\mathcal{A}_1 \subset \sigma\langle\mathcal{A}_2\rangle$ and $\mathcal{A}_2 \subset \sigma\langle\mathcal{A}_1\rangle$.

T2. Good sets and π–λ. The class of sets with the property is a σ-algebra (or a λ-system over a π-system) containing the generator.

T3. Disjointify. $B_1 = A_1$, $B_n = A_n \cap A_{n-1}^c \cap \cdots \cap A_1^c$: disjoint, same partial unions.

T4. Make it monotone. $\bigcup_n A_n = \bigcup_n D_n$ with $D_n = \bigcup_{j\le n} A_j \uparrow$; for $A_n \downarrow A$ use $A_1 \setminus A_n \uparrow A_1 \setminus A$.

T5. Squeeze. Prove measure equalities as "$\le$" and "$\ge$" separately.

T6. Uniqueness. Agreement on a generating π-system plus finiteness (Thm. 1.2.4) or σ-finiteness (Thm. 1.3.6).

T7. Nonnegative double series. $\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$ in $[0,\infty]$ when $a_{ij} \ge 0$.

T8. Common index. Finitely many members of an increasing union $\bigcup_n \mathcal{A}_n$ lie in a single $\mathcal{A}_n$; countably many need not.

T9. Null sets. Measurable subsets and countable unions of null sets are null; for a probability, pass to complements.

VII.3 Results proved on past exams

Result E1 (algebra via differences; 2016 #1(b), 2021 #1(b))**.** *$\mathcal{A} \subset \mathcal{P}(\Omega)$ is an algebra iff (i) $\Omega \in \mathcal{A}$ and (ii) $A, B \in \mathcal{A} \Rightarrow A \setminus B \in \mathcal{A}$.*

Sketch. $A \setminus B = A \cap B^c$ converts between differences and complements plus intersections.

Proof. ($\Rightarrow$) $\Omega \in \mathcal{A}$, and $A \setminus B = A \cap B^c \in \mathcal{A}$ because $\mathcal{A}$ is closed under complements and intersections. ($\Leftarrow$) $A^c = \Omega \setminus A \in \mathcal{A}$. For $A, B \in \mathcal{A}$: $A \cap B = A \setminus (A \setminus B) \in \mathcal{A}$ and $A \cup B = (A^c \cap B^c)^c \in \mathcal{A}$. So (a)–(c) of Def. 1.1.1 hold. //

Result E2 (π-system plus λ-system is a σ-algebra; 2020 #1(a))**.** *If $\mathcal{G}$ is both a π-system and a λ-system, then $\mathcal{G}$ is a σ-algebra.*

Sketch. Get an algebra from Ω, proper differences and intersections; increasing unions finish via Prop. 1.1.1.

Proof. $\Omega \in \mathcal{G}$ by λ(i). If $A \in \mathcal{G}$, then $A \subset \Omega$ gives $A^c = \Omega \setminus A \in \mathcal{G}$ by λ(ii). If $A, B \in \mathcal{G}$, then $A \cap B \in \mathcal{G}$ (π), so $A \cup B = (A^c \cap B^c)^c \in \mathcal{G}$. Hence $\mathcal{G}$ is an algebra, and it is closed under increasing unions by λ(iii); Prop. 1.1.1 makes it a σ-algebra. //

Result E3 (semialgebra plus finite disjoint unions is an algebra; 2024 #1(b))**.** *If $\mathcal{G} \neq \emptyset$ is a semialgebra closed under finite disjoint unions, then $\mathcal{G}$ is an algebra.*

Sketch. Complements are finite disjoint unions (semialgebra), so they stay in $\mathcal{G}$; then $\Omega = A \sqcup A^c$, and De Morgan gives unions.

Proof. By induction, $\mathcal{G}$ is closed under disjoint unions of any finite length. *Complements:* for $A \in \mathcal{G}$, $A^c = \bigsqcup_{i=1}^k B_i$ with disjoint $B_i \in \mathcal{G}$, so $A^c \in \mathcal{G}$. *Whole space:* pick $A \in \mathcal{G}$; then $\Omega = A \sqcup A^c \in \mathcal{G}$. *Unions:* $\mathcal{G}$ is closed under $\cap$ (semialgebra), so $A \cup B = (A^c \cap B^c)^c \in \mathcal{G}$. Hence (a)–(c) of Def. 1.1.1 hold. //

Result E4 (adding the monotone-class property gives a σ-algebra; 2024 #1(c))**.** *If $\mathcal{F} \neq \emptyset$ is a semialgebra, closed under finite disjoint unions, and a monotone class ($A_i \uparrow \Rightarrow \bigcup_i A_i \in \mathcal{F}$; $B_i \downarrow \Rightarrow \bigcap_i B_i \in \mathcal{F}$), then $\mathcal{F}$ is a σ-algebra.*

Sketch. E3 gives an algebra; an algebra closed under increasing unions is a σ-algebra.

Proof. By E3, $\mathcal{F}$ is an algebra, and it is closed under increasing unions (monotone class), so Prop. 1.1.1 applies. The decreasing-intersection property is not needed. //

Result E5 (trace σ-algebra; 2016 #2, 2020 #2(a), Problem 1.15)**.** *Let $\mathcal{F}$ be a σ-algebra on Ω and $B \subset \Omega$ (not necessarily in $\mathcal{F}$). Then $\mathcal{F}_B = \{A \cap B : A \in \mathcal{F}\}$ is a σ-algebra of subsets of B.*

Sketch. Every operation inside B is the trace of the same operation in Ω; complements are taken relative to B.

Proof. Whole space: $B = \Omega \cap B \in \mathcal{F}_B$. *Complements in B:* if $C = A \cap B$ with $A \in \mathcal{F}$, then $B \setminus C = A^c \cap B \in \mathcal{F}_B$, since $A^c \in \mathcal{F}$. *Countable unions:* if $C_i = A_i \cap B$ with $A_i \in \mathcal{F}$, then $\bigcup_i C_i = (\bigcup_i A_i) \cap B \in \mathcal{F}_B$. //

The hypotheses "$\Omega_0 \notin \mathcal{F}$" (2016) and "$B \notin \mathcal{G}$, $\mathcal{G} \neq \mathcal{P}(\Omega)$" (2020) are never used; they only rule out the easy case $B \in \mathcal{F}$, where $\mathcal{F}_B = \{A \in \mathcal{F} : A \subset B\} \subset \mathcal{F}$.

Result E6 (finitely additive and countably subadditive means measure; 2024 #1(d))**.** *Let $\mathcal{F}$ be a σ-algebra and $\lambda : \mathcal{F} \to [0, \infty]$ with $\lambda(\emptyset) = 0$, finitely additive and countably subadditive. Then λ is a measure.*

Sketch. Subadditivity gives "$\leq$"; monotonicity and finite additivity bound every partial sum, giving "$\geq$".

Proof. Monotone: $A \subset B \Rightarrow \lambda(B) = \lambda(A) + \lambda(B \setminus A) \geq \lambda(A)$. Let $A_1, A_2, \ldots \in \mathcal{F}$ be disjoint and $A = \bigsqcup_i A_i$. ($\leq$) $\lambda(A) \leq \sum_i \lambda(A_i)$ by countable subadditivity. ($\geq$) For each N, $\lambda(A) \geq \lambda(\bigsqcup_{i \leq N} A_i) = \sum_{i \leq N} \lambda(A_i)$ by monotonicity and finite additivity; let $N \to \infty$. //

Result E7 (nonnegative combinations of measures; 2020 #2(b))**.** *If μ_n are measures on $(\Omega, \mathcal{F})$ and $a_n \geq 0$ (e.g. $a_n = n^2$), then $\mu = \sum_n a_n \mu_n$ is a measure.*

Sketch. Countable additivity of each μ_n, then swap two sums of nonnegative terms.

Proof. $\mu(A) \in [0, \infty]$ and $\mu(\emptyset) = 0$. For disjoint $A_i \in \mathcal{F}$, $\mu(\bigsqcup_i A_i) = \sum_n a_n \sum_i \mu_n(A_i) = \sum_i \sum_n a_n \mu_n(A_i) = \sum_i \mu(A_i)$, where the interchange is valid because every term is nonnegative (T7). //

Result E8 (counting measure on the finite–cofinite algebra; 2020 #1(c))**.** *On $\Omega = \mathbb{N}$, counting measure $\mu(A) = |A|$ is a measure on the algebra $\mathcal{F} = \{A : A \text{ or } A^c \text{ is finite}\}$.*

Sketch. Only unions that land in $\mathcal{F}$ need checking; split on whether the union is finite or cofinite.

Proof. $\mu \geq 0$ and $\mu(\emptyset) = 0$. Let $A_n \in \mathcal{F}$ be disjoint with $A = \bigsqcup_n A_n \in \mathcal{F}$. *A finite:* only finitely many A_n are nonempty and each is finite, so $|A| = \sum_n |A_n|$. *A cofinite:* then $|A| = \infty$. If some A_n is infinite, $\sum_n |A_n| = \infty$. Otherwise infinitely many A_n are nonempty (a finite union of finite sets is finite), each contributing at least 1, so $\sum_n |A_n| = \infty$. //

Shortcut: counting measure is a measure on $\mathcal{P}(\mathbb{N})$, and restricting a measure to a sub-algebra keeps Def. 1.2.1 (a)–(c).

Result E9 (closed sets generate the Borel sets; 2020 #2(c))**.** *With $\mathcal{O}$ the open and $\mathcal{C}_{\text{cl}}$ the closed subsets of $\mathbb{R}$, $\sigma\langle\mathcal{C}_{\text{cl}}\rangle = \sigma\langle\mathcal{O}\rangle = \mathcal{B}(\mathbb{R})$.*

Sketch. Each class is the class of complements of the other, and σ-algebras are closed under complements (T1).

Proof. ($\subset$) If C is closed, $C^c \in \mathcal{O} \subset \sigma\langle\mathcal{O}\rangle$, so $C = (C^c)^c \in \sigma\langle\mathcal{O}\rangle$. Thus $\mathcal{C}_{\text{cl}} \subset \sigma\langle\mathcal{O}\rangle$, and minimality gives $\sigma\langle\mathcal{C}_{\text{cl}}\rangle \subset \sigma\langle\mathcal{O}\rangle$. ($\supset$) If O is open, then $O = (O^c)^c$ with O^c closed, so $O \in \sigma\langle\mathcal{C}_{\text{cl}}\rangle$; hence $\sigma\langle\mathcal{O}\rangle \subset \sigma\langle\mathcal{C}_{\text{cl}}\rangle$. //

Result E10 (classifying a finite class; 2020 #1(b))**.** *On $\Omega = \{a, b, c, d\}$, the class*

$$\mathcal{F} = \{\emptyset,\ \Omega,\ \{b\},\ \{a, c, d\},\ \{a, b\},\ \{c, d\}\}$$

is not a π-system, not a λ-system and not a σ-algebra.

Sketch. Exhibit one failing instance for each axiom.

Proof. Not π: $\{a, b\} \cap \{a, c, d\} = \{a\} \notin \mathcal{F}$. Not λ: $\{b\} \subset \{a, b\}$, but $\{a, b\} \setminus \{b\} = \{a\} \notin \mathcal{F}$. Not a σ-algebra (not even an algebra): $\{b\} \cup \{c, d\} \notin \mathcal{F}$, although $\mathcal{F}$ is closed under complements. //

For contrast, $\sigma\langle\mathcal{F}\rangle$ has atoms $\{a\}$, $\{b\}$, $\{c, d\}$, so it consists of the $2^3 = 8$ unions of these atoms.

Result E11 (when monotone continuity holds; 2016 #3, Problem 1.18)**.** *Take $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$ and $A_n = \{j : j \geq n\} \downarrow \emptyset$.*

- *(a) Counting measure: $\mu(A_n) = \infty$ for all n, but $\mu(\bigcap_n A_n) = \mu(\emptyset) = 0$, so m.c.f.a. fails.*
- *(b) $\mu'(A) = \sum_{j \in A} 2^{-j}$ (a probability measure): $\mu'(A_n) = 2^{-(n-1)} \to 0 = \mu'(\emptyset)$, so m.c.f.a. holds. Any finite measure works.*
- *(c) m.c.f.a. holds whenever $\mu(A_{n_0}) < \infty$ for some n_0 (Prop. 1.2.3(i)).*
- *(d) m.c.f.b. holds for every measure (Prop. 1.2.1), provided $\bigcup_n A_n \in \mathcal{F}$ when $\mathcal{F}$ is only an algebra.*

Result E12 (scaling of Lebesgue measure; 2024 #2(d))**.** *For $c \in \mathbb{R}$ and $A \in \mathcal{B}(\mathbb{R})$: $cA = \{cx : x \in A\} \in \mathcal{B}(\mathbb{R})$ and $m(cA) = |c|\, m(A)$.*

Sketch. For $c \neq 0$, $A \mapsto m(cA)$ and $A \mapsto |c|\, m(A)$ are measures on $\mathcal{B}(\mathbb{R})$ that agree on the semialgebra $\mathcal{C}$ and are σ-finite there, so Thm. 1.3.6 forces equality. Treat $c = 0$ separately.

Proof. $c = 0$. $cA = \{0\}$ (or $\emptyset$), and $m(\{0\}) = 0 = 0 \cdot m(A)$, using $0 \cdot \infty = 0$.

$c \neq 0$. (1) *cA is Borel.* $x \mapsto cx$ is a bijection of $\mathbb{R}$ mapping open sets to open sets. So $\mathcal{G} = \{A : cA \in \mathcal{B}(\mathbb{R})\}$ is a σ-algebra ($c(A^c) = (cA)^c$, $c(\bigcup_n A_n) = \bigcup_n cA_n$) containing the open sets; hence $\mathcal{B}(\mathbb{R}) \subset \mathcal{G}$ (T2).

(2) *Two measures.* $\nu_1(A) = m(cA)$ and $\nu_2(A) = |c|\, m(A)$ are measures on $\mathcal{B}(\mathbb{R})$ (disjoint A_n have disjoint images cA_n).

(3) *Agreement on $\mathcal{C}$.* For $c > 0$, $c(a, b] = (ca, cb]$; for $c < 0$, $c(a, b] = [cb, ca)$. Either way $\nu_1((a, b]) = |c|(b - a) = \nu_2((a, b])$, and $\nu_1 = \nu_2 = \infty$ on unbounded members of $\mathcal{C}$.

(4) *Uniqueness.* ν_2 is σ-finite on $\mathcal{C}$ (use $(n, n + 1]$, $n \in \mathbb{Z}$), so by Thm. 1.3.6 both ν_1 and ν_2 equal $(\nu_2|_{\mathcal{C}})^*$ on $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$. Hence $m(cA) = |c|\, m(A)$. //

Thm. 1.2.4 alone does not apply, because $m(\mathbb{R}) = \infty$. Translation invariance, $m(A + c) = m(A)$, follows from the same template, since $(a, b] + c = (a + c, b + c]$; reflection is the case $c = -1$.

Result E13 (unions and intersections of algebras; 2021 #1(c), Problems 1.4–1.5)**.** *(a) A union of two algebras need not be an algebra: on $\Omega = \{a, b, c, d\}$, $\mathcal{A}_1 = \{\emptyset, \Omega, \{a, b\}, \{c, d\}\}$ and $\mathcal{A}_2 = \{\emptyset, \Omega, \{a, c\}, \{b, d\}\}$ are algebras, but $\mathcal{A}_1 \cup \mathcal{A}_2$ is not. (b) Any intersection $\bigcap_{\theta \in \Theta} \mathcal{F}_\theta$ of σ-algebras on Ω is a σ-algebra.*

Sketch. (a) Each $\mathcal{A}_i$ has the form $\{\emptyset, B, B^c, \Omega\}$; for the union, exhibit one pair whose union is missing. (b) Each axiom holds in every $\mathcal{F}_\theta$, hence in the intersection.

Proof. (a) $\mathcal{A}_1 = \{\emptyset, B, B^c, \Omega\}$ with $B = \{a, b\}$ contains Ω, is closed under complements, and is closed under pairwise unions ($C \cup \emptyset = C$, $C \cup \Omega = \Omega$, $B \cup B^c = \Omega$). The same holds for $\mathcal{A}_2$ with $B = \{a, c\}$. But $\{a, b\}, \{a, c\} \in \mathcal{A}_1 \cup \mathcal{A}_2$, while $\{a, b\} \cup \{a, c\} = \{a, b, c\} \notin \mathcal{A}_1 \cup \mathcal{A}_2$.

(b) Let $\mathcal{F} = \bigcap_\theta \mathcal{F}_\theta$. $\Omega \in \mathcal{F}_\theta$ for all θ, so $\Omega \in \mathcal{F}$. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}_\theta$ for all θ, so $A^c \in \mathcal{F}$. If $A_n \in \mathcal{F}$, then $\bigcup_n A_n \in \mathcal{F}_\theta$ for all θ, so $\bigcup_n A_n \in \mathcal{F}$. //

Problem 1.5 is the analogue on $\{1, 2, 3\}$: $\{\emptyset, \Omega, \{1\}, \{2, 3\}\} \cup \{\emptyset, \Omega, \{1, 2\}, \{3\}\}$ misses $\{1\} \cup \{3\}$. Since Ω is finite, $\mathcal{A}_1$ and $\mathcal{A}_2$ are also σ-algebras, and $\mathcal{A}_1 \cup \mathcal{A}_2$ is still a λ-system (Section VIII).

Result E14 (increasing unions of algebras; 2021 #1(d))**.** *If $\mathcal{A}_1 \subset \mathcal{A}_2 \subset \cdots$ are algebras on Ω, then $\mathcal{A} = \bigcup_{n \geq 1} \mathcal{A}_n$ is an algebra.*

Sketch. Because the $\mathcal{A}_n$ increase, finitely many members of $\mathcal{A}$ lie in a common $\mathcal{A}_n$ (T8); apply the algebra axioms there.

Proof. $\Omega \in \mathcal{A}_1 \subset \mathcal{A}$. If $A \in \mathcal{A}$, then $A \in \mathcal{A}_{n_0}$ for some n_0, so $A^c \in \mathcal{A}_{n_0} \subset \mathcal{A}$. If $A, B \in \mathcal{A}$, say $A \in \mathcal{A}_{n_1}$ and $B \in \mathcal{A}_{n_2}$, then $A, B \in \mathcal{A}_n$ for $n = \max(n_1, n_2)$, so $A \cup B \in \mathcal{A}_n \subset \mathcal{A}$. //

The σ-version fails: a sequence $A_k \in \mathcal{A}_{n_k}$ with $n_k \to \infty$ need not lie in one $\mathcal{A}_n$. On $\Omega = \mathbb{N}$, the σ-algebras $\mathcal{F}_n = \sigma\langle\{1\}, \ldots, \{n\}\rangle = \{A : A \subset \{1, \ldots, n\}$ or $A^c \subset \{1, \ldots, n\}\}$ increase to the finite–cofinite algebra, which is not a σ-algebra (E17).

Result E15 (sets that differ by a null set; 2021 #2(a)). *Let μ be a measure on $(\Omega, \mathcal{F})$ and $A, B \in \mathcal{F}$. If $\mu(A \triangle B) = 0$, then $\mu(A) = \mu(B)$.*

Sketch. $A \setminus B$ and $B \setminus A$ are null by monotonicity, so $\mu(A)$ and $\mu(B)$ both equal $\mu(A \cap B)$.

Proof. Since $A \setminus B \subset A \triangle B$, monotonicity (Prop. 1.2.2(i)) gives $\mu(A \setminus B) = 0$. By finite additivity, $\mu(A) = \mu(A \cap B) + \mu(A \setminus B) = \mu(A \cap B)$. Symmetrically, $\mu(B) = \mu(A \cap B)$. //

Nothing is subtracted, so no finiteness is needed: this holds for every measure, not only for P.

Result E16 (sets of probability one; 2021 #2(b)). *Let P be a probability measure on $(\Omega, \mathcal{F})$. If $B_i \in \mathcal{F}$ and $P(B_i) = 1$ for all $i \geq 1$, then $P(\bigcap_i B_i) = 1$.*

Sketch. Pass to complements: a countable union of null sets is null by countable subadditivity; then De Morgan.

Proof. $P(B_i^c) = 1 - P(B_i) = 0$ for each i. By De Morgan and countable subadditivity (Prop. 1.2.3(ii)), $P\big((\bigcap_i B_i)^c\big) = P\big(\bigcup_i B_i^c\big) \leq \sum_i P(B_i^c) = 0$, so $P(\bigcap_i B_i) = 1 - 0 = 1$. //

For a general measure, the same argument shows that a countable union of null sets is null.

VII.4 Some potential next questions

Result E17 (finite–cofinite versus countable–cocountable; Exs. 1.1.3, 1.1.4, 1.1.7). *Let $|\Omega| = \infty$. (a) $\mathcal{F}_5 = \{A : A \text{ or } A^c \text{ finite}\}$ is an algebra but not a σ-algebra. (b) $\mathcal{F}_6 = \{A : A \text{ or } A^c \text{ countable}\}$ is a σ-algebra. (c) $\sigma\langle\mathcal{F}_5\rangle = \mathcal{F}_6$.*

Sketch. Both classes are symmetric under complements, so check intersections or unions case by case; for (c) use two containments.

Proof. (a) $\Omega \in \mathcal{F}_5$ and $\mathcal{F}_5$ is closed under complements by symmetry. If A or B is finite, so is $A \cap B$; if A^c and B^c are finite, so is $(A \cap B)^c = A^c \cup B^c$. So $\mathcal{F}_5$ is an algebra. For distinct $\omega_1, \omega_2, \ldots$, the set $\{\omega_1, \omega_3, \ldots\}$ is a countable union of singletons in $\mathcal{F}_5$, but neither it nor its complement is finite.
(b) $\Omega \in \mathcal{F}_6$, and $\mathcal{F}_6$ is closed under complements by symmetry. If every A_n is countable, so is $\bigcup_n A_n$; if some $A_{n_0}^c$ is countable, then $(\bigcup_n A_n)^c \subset A_{n_0}^c$ is countable.
(c) ($\subset$) $\mathcal{F}_5 \subset \mathcal{F}_6$ and $\mathcal{F}_6$ is a σ-algebra, so $\sigma\langle\mathcal{F}_5\rangle \subset \mathcal{F}_6$. ($\supset$) A countable A is a countable union of singletons, which lie in $\mathcal{F}_5$, so $A \in \sigma\langle\mathcal{F}_5\rangle$; if A^c is countable, then $A^c \in \sigma\langle\mathcal{F}_5\rangle$, and so $A \in \sigma\langle\mathcal{F}_5\rangle$. //

VIII Examples, counterexamples and a self-check

Phenomenon	Example	Where
Algebra, not a σ-algebra	finite–cofinite sets of an infinite Ω	Ex. 1.1.3, E17
Union of σ-algebras not an algebra	$\{\emptyset, \Omega, \{a,b\}, \{c,d\}\} \cup \{\emptyset, \Omega, \{a,c\}, \{b,d\}\}$ on $\{a,b,c,d\}$	2021 #1(c), E13
λ-system, not a π-system	the same union: $\{a,b\} \cap \{a,c\} = \{a\}$ is missing	E13

Phenomenon	Example	Where
Increasing union of σ-algebras, not a σ-algebra	$\sigma\langle\{1\},\dots,\{n\}\rangle \uparrow$ finite–cofinite sets of $\mathbb{N}$	E14
Algebra, not a λ-system	finite–cofinite sets of $\mathbb{N}$: $\{2,4,\dots,2n\} \uparrow$ the evens	Ex. 1.1.9
Semialgebra, not an algebra	$\{(a,b],(a,\infty)\}$ on $\mathbb{R}$; all intervals	Exs. 1.3.1–1.3.2
Contains every singleton, still not $\mathcal{B}(\mathbb{R})$	$\sigma\langle\text{singletons}\rangle = \mathcal{F}_6$	Prob. 1.12
m.c.f.a. fails	counting measure on $\mathbb{N}$, $A_n = \{j \geq n\} \downarrow \emptyset$	Prob. 1.18, E11
Not σ-finite	counting measure on an uncountable Ω	Ex. 1.2.1
Positive mass at a point	$\mu_F(\{x\}) = F(x+) - F(x-)$ at a jump of F	§1.3.2(i)
Uncountable null set	the Cantor set	§1.3.2(iii)
Incomplete measure space	$(\mathbb{R},\mathcal{B}(\mathbb{R}),m)$; $(\mathbb{R},\mathcal{B}(\mathbb{R}),\text{point mass})$	§1.4
Every subset measurable	discrete F: $\mathcal{M}_{\mu_F^*} = \mathcal{P}(\mathbb{R})$	Prob. 1.28
Non-measurable set	Vitali set (axiom of choice)	§1.3.2(v)

Self-check

Question	Answer
Fastest ways to prove "σ-algebra"?	Algebra + increasing unions (Prop. 1.1.1); π-system + λ-system (E2); an intersection of σ-algebras (E13).
How does a semialgebra become an algebra?	Add finite disjoint unions: $\mathcal{A}(\mathcal{C})$ is the smallest algebra containing $\mathcal{C}$ (Prob. 1.19, E3).
Union of two algebras? Increasing union of algebras? Of σ-algebras?	Need not be an algebra (E13); an algebra (E14); need not be a σ-algebra (E14).
Does m.c.f.b. always hold? m.c.f.a.?	m.c.f.b. yes, for every measure; m.c.f.a. needs $\mu(A_{n_0}) < \infty$ for some n_0.
What does the π–λ theorem give?	$\lambda\langle\mathcal{C}\rangle = \sigma\langle\mathcal{C}\rangle$ for a π-system $\mathcal{C}$, so a λ-system containing $\mathcal{C}$ contains $\sigma\langle\mathcal{C}\rangle$.
$\mu(A \triangle B) = 0$: what follows?	$\mu(A) = \mu(A \cap B) = \mu(B)$, for every measure (E15).
Countably many sets of probability one?	Their intersection has probability one: the complements are null, and a countable union of null sets is null (E16).
Hypotheses of Thm. 1.2.4?	Finite μ_1, μ_2; equal on a π-system $\mathcal{C}$ with $\sigma\langle\mathcal{C}\rangle = \mathcal{F}$; equal total mass.
Uniqueness for Lebesgue measure?	Thm. 1.3.6 (σ-finite on the semialgebra); Thm. 1.2.4 needs finiteness.
Is μ^* a measure on $\mathcal{P}(\Omega)$?	No: only an outer measure (monotone, countably subadditive); it is a measure on $\mathcal{M}_{\mu^*}$.
What does Carathéodory produce?	A σ-algebra $\mathcal{M}_{\mu^*} \supset \sigma\langle\mathcal{C}\rangle$ and a complete measure μ^* on it that extends μ.
Why half-open intervals in $\mathcal{C}$?	Their complements are finite disjoint unions of such intervals, and $\mu_F((a,b]) = F(b) - F(a)$ is additive.
$\mu_F^*(\{x\})$? $m(\mathbb{Q})$?	$F(x+) - F(x-)$, the jump at x; $m(\mathbb{Q}) = 0$ since countable sets are null.
Is $(\mathbb{R},\mathcal{B}(\mathbb{R}),m)$ complete?	No; its completion is $(\mathbb{R},\mathcal{M}_m,m)$.
Probability measures on $\mathcal{B}(\mathbb{R})$ versus cdfs?	One-to-one: $P \mapsto F(x) = P((-\infty,x])$, and $P = \mu_F^*$ on $\mathcal{B}(\mathbb{R})$.

IX Appendix A tools that Chapter 1 uses

- **De Morgan (Prop. A.1.1).** $(\bigcup_\alpha A_\alpha)^c = \bigcap_\alpha A_\alpha^c$ and $(\bigcap_\alpha A_\alpha)^c = \bigcup_\alpha A_\alpha^c$, for any index set.
- **Countability (Def. A.1.11, Problems A.5–A.6).** Countable unions of countable sets are countable; $\mathbb{Q}$ and $\mathbb{N}^k$ are countable; $\{0,1\}^{\mathbb{N}}$, $\mathcal{P}(\mathbb{N})$ and $\mathbb{R}$ are not.
- **Axiom of choice (Def. A.1.8).** Products of nonempty sets are nonempty; needed for non-measurable sets.
- **Infimum (Def. A.2.1).** If $s = \inf S$ is finite, every $\epsilon > 0$ has some $x \in S$ with $x < s + \epsilon$ (near-optimal covers for μ^*).
- **Density of $\mathbb{Q}$ (Cor. A.2.2).** Between any two reals lies a rational (rational generators of $\mathcal{B}(\mathbb{R})$).
- **Series (Def. A.2.5, Ex. A.2.1).** Nonnegative series converge in $[0,\infty]$ and may be rearranged freely; $\sum_{j\geq 0} r^j = 1/(1-r)$ for $0 < r < 1$.
- **Open, closed, compact (Defs. A.4.8–A.4.10).** Closed means the complement is open; compact means every open cover has a finite subcover.
- **Heine–Borel (Thm. A.4.2).** $K \subset \mathbb{R}$ is compact iff it is closed and bounded (used for countable additivity of μ_F on $\mathcal{C}$).